

Bethel Melesse Tessema

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Dear Hiring Manager,

I'm an **AI/ML engineer and researcher** with a **master's degree in Artificial Intelligence** and more than **three years of experience** across research and industry. I am also the founder of **PARAHIGG**, an AI-powered legal research platform working to make Ethiopian legislation and case law more searchable and easier to understand.

My work sits at the intersection of **deep learning**, language, and scientific computing. I have built end-to-end machine learning pipelines across **natural language processing, computer vision**, and **computational drug discovery**, with particular strength in large-scale data processing, model evaluation, and reproducible experimentation.

During my research at **Ajou University's DBDC Lab**, I explored how AI can better serve **low-resource languages**. This included training transformer-based retrieval models for African languages, improving zero-shot generalization in multimodal classifiers, and creating monolingual datasets for seven low-resource languages from more than **1,000 TB** of Common Crawl data at a remarkably low cost. That work led to **UnifiedCrawl**, a publication on affordable LLM adaptation for low-resource languages.

In industry, I worked as an **AI Engineer at 3Billion**, where I applied machine learning to **structure-based drug discovery**. I developed models for evaluating the synthetic feasibility of molecular reaction pathways and contributed to an end-to-end protein-ligand interaction prediction system. These projects combined **graph neural networks**, molecular data processing, **distributed multi-GPU training**, and close validation with experimental chemists.

PARAHIGG brings together my interest in language technology, accessible information, and building useful products. I am developing it from the ground up—from collecting and processing legal documents to retrieval and **LLM-based question answering**—so that Ethiopian legal information can be more accessible to lawyers, students, and the wider public.

My path has taken me from software engineering studies in Ethiopia to international studies and AI research in **South Korea**. Beyond technical work, I have volunteered with NGOs and supported the Korean War Veterans Revisit Program as a translator and interpreter. Those experiences continue to shape how I approach technology: as something that should connect people, lower barriers, and create practical value.

Sincerely,

Bethel Melesse Tessema